5. Control of Ionizing Radiation

Identification of Radiation Safety Problems

Ionizing radiation cannot be felt, seen, heard, tasted, or smelled. This gives it a mysterious quality. However, unknown radioactive isotopes can be identified and measured with the instruments described in the last chapter. Ionizing radiation can be adequately monitored and properly controlled.

Areas where radiation dose rates may be excessive can be guarded during exposure time by suitable methods: erecting barriers and warning signs, stationing attendants to keep personnel out of restricted localities and, in extreme cases, completely closing the areas. Safe exposure times and safe, operating practices can be established through measurement and control and by profiting from experience.

Work with radiation can and should be so planned and managed that radiation exposure of employees and the general public is kept to a minimum. No radioactive material that can result in exposures above the established levels should be released through contaminated air or water, through loss of control over contaminated waste materials, equipment or shipments, or by employees unknowingly leaving the place of work with contamination of their persons, clothing, or shoes.

When examining the work environment for a potential radiation safety problem, many factors must be considered. First, the type of work in which the employee is involved must be evaluated. The potential level of exposure for working near the processing of feed materials for nuclear reactors is much greater that that which would be present in the area of an alpha-emitting static eliminator or beta ray thickness gauge. Further, radiation fields occurring with X-ray machines can be much more controlled than an unsealed radioisotope source. It is also dependent upon how involved the employee is with the process. For example, employees involved with the application of radioactive compounds, such as that applied to luminous instrument dials, would be in much closer contact with the radioactive source than an individual who is monitoring radioactive tracers used as labels in large pipeline distribution systems.

The second consideration that must be made is the source of the radiation. The amount and type of radiation is important. As mentioned, those radioactive sources that are instruments or machines can be turned off, whereas radioisotopes will emit radiation constantly. Also, each radionuclide has a radio toxicity per unit activity that varies from nuclide to nuclide.

Further, sealed sources are not nearly so dangerous as unsealed sources. Finally, if the radioisotope is in the form of a radioactive metal that is going through some type of machining process, a major problem will be the spread of loose material--e.g., flaking, grinding chips--and the contamination of the employee.

If the operational factors are considered, the required level of radiation protection and potential problems that might arise can be determined. The operational factors that should be evaluated include such things as:

1. The area involved (square feet), number of rooms and buildings.
2. Number of employees potentially exposed to the radiation and the location of those employees.
3. The chemical and physical states of the radioactive material and its use.
4. Potential incidents that are likely to occur that would increase the potential exposure to the employees.
5. Nonradiation hazards that might be involved; that is, high voltage, toxicity of chemical substances.
6. The nature of the probable exposure to the employees; that is, whether or not the exposure would be a controlled or supervised release such as through some type of disposal procedure, accidental release that may not be sensed by warning devices, or a violent release of dust droplets or gases--such as through fire or explosion--that may carry radioactive contamination.
7. The inherent danger of the materials and procedure being implemented.
8. The probability of detection of a potentially harmful radioactive situation by routine surveys and monitoring.
9. Possible effects of a radiation accident on operation; for example, loss of production, loss of space, and cleanup costs.

Finally, consideration must be given to the potential employee exposure. Efforts should be made to calculate the maximum dosage, given a specific type of incident, that the employee might receive. This should be done for both external and internal radiation potentials.

If the above factors are given consideration in the development of an overall radiation safety analysis, then potential employee exposure will be minimized.

Authorization for Radionuclide Use

Generally, the use of radionuclides and radiation-producing instruments (X-ray machines) is licensed through the Federal government or state health

departments. The Nuclear Regulatory Commission (NRC) has established guidelines for the licensing of radionuclide users. Table 6.5.1 indicates the minimum quantity for radioactive materials that must be licensed through the NRC. Because of the variation in state requirements, licensing procedures should be investigated if radionuclide sources are to be used.

In general, any request for license requires the following information: the type and potential use of radionuclide along with the maximum quantities; the training and experience of the users; the laboratory facilities, handling equipment, and the monitoring procedures; disposal procedures, if necessary; and an outlined radiation-protection program.

Although the above requirements need to be met only if NRC licensure is being sought, these requirements for licensure provide a good general outline of the components of a radiation safety program.

Table 6.5.1

Sample licensure.

Radionuclide	Minimum Quantity for Radioactive Materials Sign in Room (μCi)	Minimum Quantity for Radioactive Materials Label (μCi)
Calcium-25	100	10
Carbon-14	1,000	100
Cesium-137 and Barium-137	100	10
Chlorine-36	100	10
Chromium	10,000	1000
Cobalt-60	10	1
Copper-64	1,000	100
Gold-198	100	100
Hydrogen-3	10,000	1000
Iodine-125	10	1
Iodine-131	10	1
Iron-55	1,000	100
Iron-59	100	10
Phosphorus-32	100	10
Potassium-42	100	10
Radium-226	0.1	0.01
Sodium-24	100	10
Strontium-90 and Yttrium-90	1	0.1
Zinc-65	100	10
Unidentified, but not α emitter	1	0.1
Unidentified α emitter	0.1	0.01

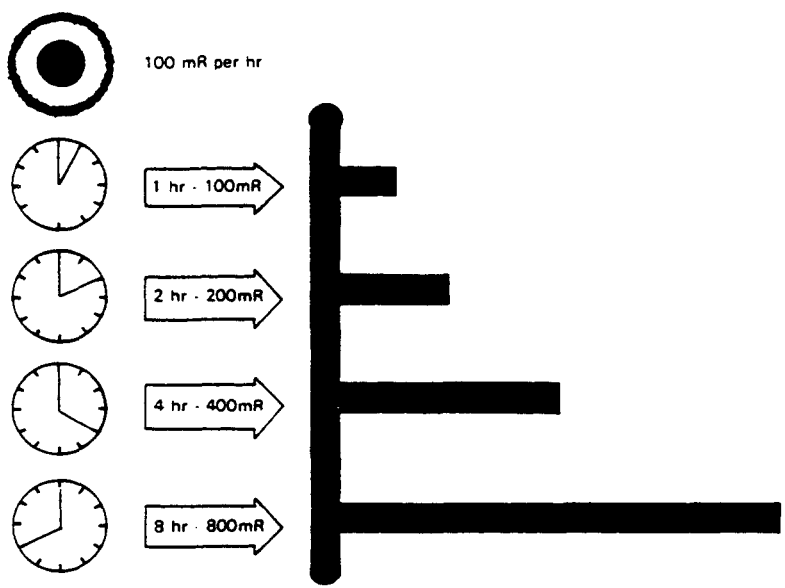
Protection from Radiation Hazard

There are three basic tools that can provide protection against a radiation source. These are time, distance, and shielding. The goal of the protection is to prevent overexposure from external radiation and to minimize the entry of radionuclides into the body, or minimize internal radiation.

Time. As an element of protection, time is almost self-explanatory. Radiation occurs at a rate of roentgens (or rads) per hour. Therefore, the shorter the time of exposure, the smaller the radiation dose received by the personnel. Figure 6.5.1 illustrates this concept.

Figure 6.5.1

Time vs. radiation exposure.



According to the figure, if someone is in an area where the radiation level from penetrating external radiation is 100 mR/hour, then in one hour 100 mR would be received. After two hours, the exposure dose would be 200 mR. Therefore, if work procedures can be reviewed and analyzed such that the time of exposure is minimized, then the exposure dose received by the employee is also minimized. Time also plays an important role with respect to protection when considering the factor of "half-life." If an exposure potential is high in a given area and the radioactive source has a short half-life, it is possible to just wait for radioactive decay to occur; and the activity of the radionuclide will be reduced. As can be seen in Figure 6.3.5, at the end of

seven half-lives the activity is reduced to one percent of the original activity. It is possible to calculate the loss of activity due to radioactive decay if the original activity and date of measure are known.

The decrease of activity is equal to $(1/2)^n$, where n is equal to the number of half-lives that have passed since the last known measurement.

Example

An isotope user acquired from a national laboratory a surplus Co-60 source in June, 1977. The source was last measured for activity in January, 1970. At that time, the source had an activity of 40 mR/hr at 10 cm.
What is the anticipated exposure rate at 10 cm in June, 1977?

Solution

January 1970 to June 1977 = 7.5 years
Half-life of Co-60 = 5.3 years
Half-life transpired = 7.5/5.3 = 1.415 half lives
Using $(1/2)^n$ $(1/2)^{1.415}$ = 0.375

$$\text{Exposure rate (June 1977)} = \frac{40 \text{ mR}}{\text{hr}} \times 0.375 = \frac{15 \text{ mR}}{\text{hr}}$$

For convenience, tables for $(1/2)^n$ have been developed for use as illustrated in Table 6.5.2.

Table 6.5.2

Powers of one-half for attenuation and decay calculations, $(1/2)^{n*}$.

n	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	1.000	0.933	0.871	0.812	0.758	0.707	0.660	0.616	0.578	0.536
1	0.500	0.467	0.435	0.406	0.379	0.354	0.330	0.308	0.287	0.268
2	0.250	0.233	0.217	0.203	0.190	0.177	0.165	0.154	0.144	0.134
3	0.125	0.177	0.109	0.102	0.095	0.088	0.083	0.077	0.072	0.067
4	0.063	0.058	0.054	0.051	0.047	0.044	0.041	0.039	0.036	0.034
5	0.031	0.029	0.027	0.025	0.024	0.022	0.021	0.019	0.018	0.017
6	0.016	0.015	0.014	0.013	0.012	0.011	0.010			

* "n" is the number of half-value layers or half-lives.

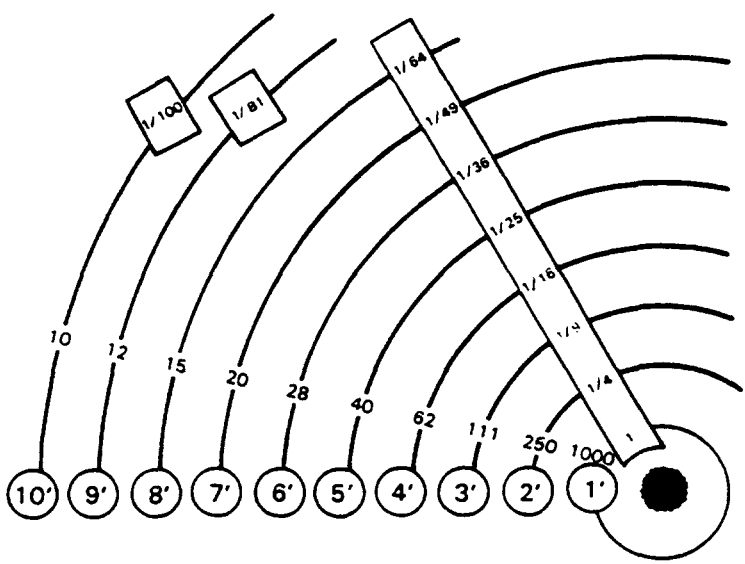
Distance. The second tool used for protection against radiation exposure is distance. The exposure rate is reduced by a factor of one divided by the square of the distance between the employee and the source. This is referred to as the "inverse-square law." Figure 6.5.2 demonstrates the effect of the inverse-square law on the intensity of radiation.

If there is a point source of radiation giving off 1000 units of penetrating external radiation at one foot, a worker would receive only one-fourth as much, or $(1/2)^2$, if the distance is doubled to two feet. If the distance is tripled, the dose is reduced to one-ninth, or $(1/3)^2$.

For example, if it is known that the radiation level is 0.1 R/hr at one foot, what would the exposure be if the worker was moved to a distance of five feet? Using the inverse-square law, the radiation level at five feet would be equal to 1.0 R/hr divided by $(5)^2$, or 0.04 R/hr. The advantage of distance as a protective tool when working with radiation can be seen from this example. The example illustrates the valuable use of tools such as long tongs. One important point that must be mentioned is that the distance of measurement and the units used in the "inverse-square law" must be equivalent; that is, if exposure levels are determined at X feet, the variations in distance must also be calculated in feet.

Figure 6.5.2

Distance vs. radiation exposure.



Shielding. The third tool to be used in protection against radiation is shielding. Shielding reduces radiation to the employee by placing a radiation-absorbing barrier between the source and the employee to be protected. The selection of the shielding is dependent upon the type and quantity of the radionuclide present. Table 6.5.3 illustrates some typical shielding materials used for the various types of radiation. The arrangement of the materials in the table is in the general order of increased thickness required to be an adequate shield.

Table 6.5.3

Typical shielding materials for radionuclides.

	Shielding Material		
	Permanent	Temporary	Additional Clothing
Alpha	Unnecessary	Unnecessary	Unnecessary
Beta	Lead, copper, iron, aluminum	Iron, aluminum, plastics, wood	Leather, rubber, plastic, cloth
Gamma, X-rays	Lead, iron, copper, lead glass, heavy aggregate concrete, aluminum, ordinary concrete, plate glass, wood, water, paraffin	Lead, iron, lead glass, aluminum, concrete blocks, water, wood	Lead fabrics (but not for "hard" gamma)

Care must be taken to ensure that the location and shape of the shield limits radiation in all directions which may provide potential exposure to all employees; e.g., floor, ceiling. Thickness of the shielding required is relatively independent of the distance from the source or object. However, to minimize the size of the shield and to maximize protection, the shield should be as close to the source as possible as illustrated in Figure 6.5.3.

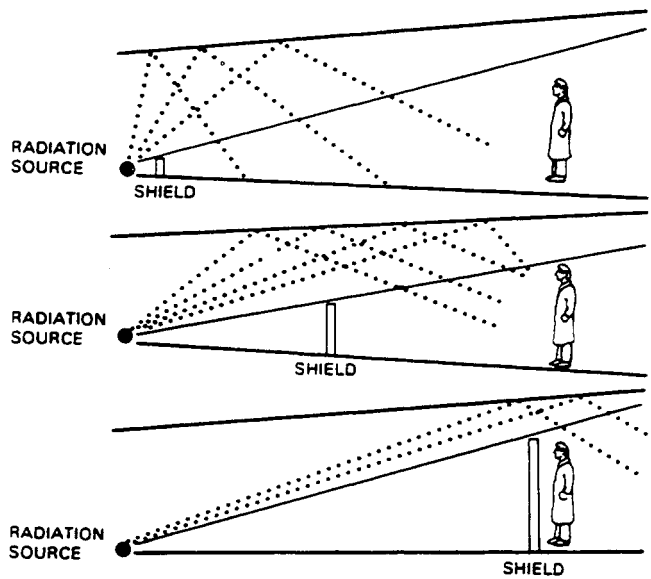
By placing the shield in close proximity to the source, incident radiation is minimized, and the expenditure for the shield is also reduced. Shields may be cylindrical or box shaped, depending upon local needs. Local shields are often built of lead bricks or concrete blocks, as presented in Figure 6.5.4.

A major concern in developing local shielding with any type of block or brick is that radiation may leak through the cracks in the blocks, or incident radiation may be reflected off walls or even the shield itself. In many instances, the radioactive source is so active that the shielding limits the employee to approaching the radioactive source from a distance or by remote control (Figure 6.5.5).

Gamma Radiation Shielding. The purpose of shielding is to reduce the radiation exposure to an acceptable level; the shield absorbs the radiation. Gamma radiation is photon energy with a high penetrating capability; and,

Figure 6.5.3

Shielding placement.



REDUCTION OF SCATTER BY PLACEMENT OF SHIELD CLOSE TO EITHER SOURCE OR PERSON TO BE SHIELDED

therefore, materials of high density--including lead, iron, and concrete--are used as shielding materials. The absorption of gamma radiation by a shield is an exponential function related by the equation

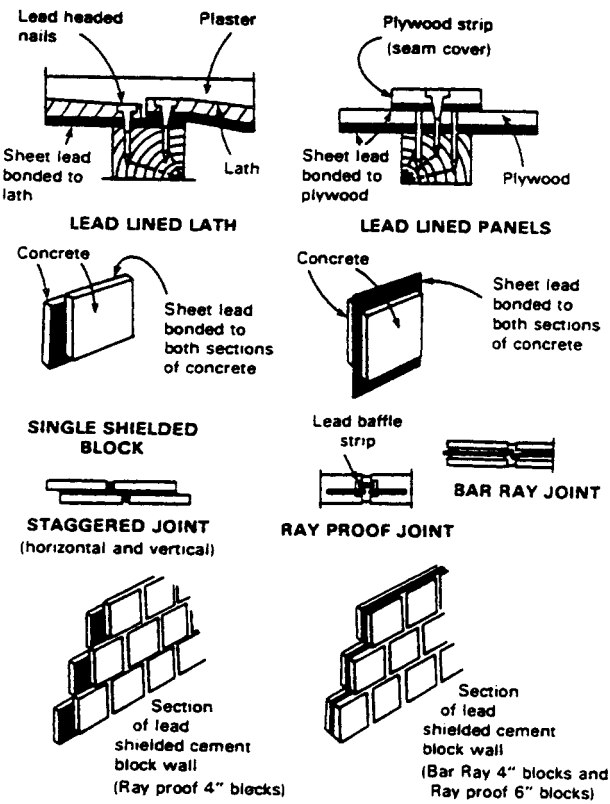
$$I = I_0 e^{-\mu x}$$

where I is the exposure rate (R/hr) after passing through the shield, I_0 is the initial exposure rate (R/hr) at the shield, μ is the absorption coefficient of the shielding materials (cm^{-1}), and x is the shield thickness (cm). The absorption coefficient is a function of the energy level of the radiation and the properties of the shielding materials. Often the absorption coefficient is expressed as a mass attenuation coefficient (cm^2/g) (Table 6.5.4). To calculate μ, given the mass attenuation coefficient, it is necessary to divide the mass attenuation coefficient by the density of the shielding material. Therefore, shielding requirements or effects of a given shield can be calculated.

The effect of shielding is also often expressed in terms of half value layers (HVL). The half value layer is the thickness of a specified shielding material to reduce the exposure rate from a specific radiation source by a factor of 0.5. For example, in 2.6 half value layers, the number of photons will be reduced $(0.5)^{2.6}$ or 0.165 of the original number entering the

Figure 6.5.4

Materials for shielding.



specified shield. Using $(0.5)^n$ or Table 6.5.2, it is possible to calculate the number of half value layers required for a desired attenuation of the radiation. It also allows for the calculation of a given attenuation for a certain thickness of a shield. Table 6.5.5 illustrates some typical half value layers for five isotopes. The calculation of HVL is similar to the calculation of the effect of half life: i.e., attenuation shield = $(0.5)^{HVL}$.

An example may further explain the shielding calculation.

Example

Calculate the thickness of a lead shield necessary to reduce the exposure rate from 10.0 R/hr to 2.5 R/hr from a Cc-60 source.

Figure 6.5.5

Master-slave manipulation.

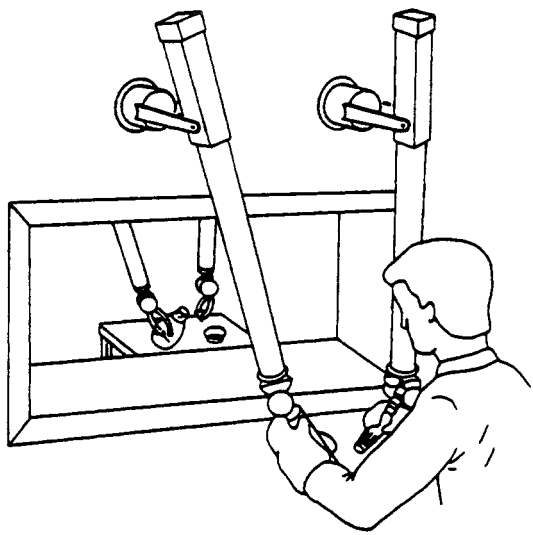


Table 6.5.4

Mass attenuation coefficients.

Photon Energy MeV	Mass Attenuation Coefficient in cm^2/g for--				
	Aluminum	Iron	Lead	Water	Concrete
0.01	26.3	173	133	5.18	26.9
0.02	3.41	25.5	85.7	0.775	3.59
0.05	0.369	1.94	7.81	0.227	0.392
0.1	0.171	0.370	5.40	0.171	0.179
0.5	0.0844	0.0840	0.161	0.0968	0.087
1.0	0.0613	0.0599	0.0708	0.0707	0.0637
5.0	0.0284	0.0314	0.0424	0.0303	0.0290
10.0	0.0231	0.0298	0.0484	0.0222	0.0231

Source: "Radiological Health Handbook," U.S. DHEW, Public Health Service, 1970.

Table 6.5.5

Data for gamma-ray sources.

Radio-isotope	Atomic Number	Half Life	Gamma Energy (MeV)	Half-Value Layer† Conc. in	Steel in	Lead cm	Tenth-Value Layer† Conc. in	Steel in	Lead cm	Specific Gamma-Ray Constant R cm^3/m Ci-h*
Cesium-137	55	27 y	0.66	1.9	0.64	0.65	6.2	2.1	2.1	3.2
Cobalt-60	27	5.24 y	1.17, 1.33	2.6	0.82	1.20	8.2	2.7	4.0	13.0
Gold-198	79	2.7 d	0.41	1.6	--	0.33	5.3	--	1.1	2.32
Iridium-192	77	74 d	0.13 to 1.06	1.7	0.50	0.60	5.8	1.7	2.0	5.0¶
Radium-226	38	1622 y	0.047 to 2.4	2.7	0.88	1.66	9.2	2.9	5.5	8.25§

From "NCRP Report No. 34," National Council on Radiation Protection and Measurements, Washington, D.C., 1971.

†Approximate values obtained with large attenuation.
*These values assume that gamma absorption source is negligible. Value is R/millicurie-hour at 1 cm can be converted to R/Ci-h at 1 meter by multiplying the number in this column by 0.10.
¶This value is uncertain.
§This value assumes that the source is sealed within 0.5 mm thick platinum capsule, with units of R/mgh at 1 cm.

Solution

HVL Method

$$\text{Attenuation required} = \frac{2.5 \text{ R/hr}}{10.0 \text{ R/hr}} = 0.25$$

Calculate HVL necessary

$$(0.5)^{HVL} = 0.25$$

$$HVL = \frac{\log 0.25}{\log 0.5} = \frac{-0.602}{-0.301} = 2 \text{ HVL}$$

$$2 \text{ HVL} \times \frac{1.25 \text{ cm Pb}}{\text{HVL}} = 2.5 \text{ cn Pb}$$

I/I_0 Method

$I = I_0 e^{-\mu x}$

$x = -\ln(I/I_0)/\mu$

$I = 2.5$ R/hr

$I_0 = 10.0$ R/hr

Using table

$\mu = 0.060 \text{ cm}^2/\text{g} \times 11.34 \text{ g/cm}^3 = 0.680 \text{ cm}^{-1}$

Note: Pb (density) = 11.34 g/cm^3

$$x = \frac{-\ln(2.5/10.0)}{0.68 \text{ cm}^{-1}} = 2.04 \text{ cm}$$

Example

An unshielded source of Cesium-137 has an exposure rate of 0.1 mR/hr. What would be the exposure rate if a 3 centimeter lead shield were placed around the source? (HVL: Cesium-137 = 0.64 cm lead (Pb), MeV: Cs-137 = 0.66 MeV)

Solution

HVL Method

The solution to this example is rather straightforward. The HVL of these gamma photons is 0.65 cm. The number of HVL in a 3 cm shield would then be equal to 3.0 0.65 or 4.62 half value layers. Using Table 6.5.2, n is equal to 4.62, which when extrapolated would indicate an attenuation coefficient of approximately 0.41. Therefore, the exposure rate with a 3 cm lead shield surrounding the source would be equal to 0.1 mR/hr x 0.041 or 4.1 μR/hr.

I/I_o Method

$I = I_o e^{-\mu x}$
$I = ?$
$I_o = 0.1$ mR/hr
$X = 3$ cm
$\mu = 0.105\ cm^2/g \times 11.34\ g/cm^3 = 1.19\ cm^{-1}$ (from table)
$I = 0.1\ mR/hr \times e^{-(1.19 \times 3)}$
$I = 2.82$ R/hr

In the previous example, the calculated values as determined by the two methods are somewhat different. More important, the effect as calculated by the I/I_o method is significantly less. This is because the formula, $I = I_o e^{-\mu x}$, does not take into consideration a "buildup" phenomenon. In matter being traversed by a beam of electromagnetic radiation, there is actually a higher intensity of photons at any point in the matter than would be predicted by the equation. This is true because of the presence of X-radiation and secondary or scattered photons produced as the radiation penetrates the matter. The increase in exposure rate, buildup (B), is not easily calculated but must be considered when determining shielding requirementsN A factor B is included in the attenuation equation $I = BI_o e^{-\mu x}$, with the tabulated values of B for specific conditions available in the literature. Table 6.5.6 illustrates the dose buildup factor for a point isotropic source. Using such a table, the factor B can be included in the shielding calculation.

Example

Calculate the exposure rate (I) for the previous example allowing for the buildup factor.

Solution

$I = BI_o e^{-\mu x}$
$I_o = 0.1$ mR/hr
$\mu = 0.105\ cm^2/g \times 11.34\ g/cm^3 = 1.19\ cm^{-1}$
$x = 3.0$ cm Pb

Using table 6.5.6 for

$\mu x = 1.19\ cm^{-1} \times 3.0\ cm = 3.57$
MeV for Cs-137 = 0.66 MeV
Interpolating, B = 1.79
$I = 1.78 \times 0.1\ mR/hr \times e^{-(3.57)}$
$I = 5.04\ \mu R/hr$

One final calculation that can be made when examining gamma radiation is that of projecting the potential exposure rate at a given distance of a gamma source. To make this calculation, it is necessary to know the type of radionuclide, the relative energies of that nuclide, and its activity. The

Table 6.5.6

Dose buildup factor (B) for a point isotropic source.

Material	MeV	μX*						
		1	2	4	7	10	15	20
Water	0.255	3.09	7.14	23.0	72.9	166	456	982
	0.5	2.52	5.14	14.3	38.8	77.6	178	334
	1.0	2.13	3.71	7.68	16.2	27.1	50.4	82.2
	2.0	1.83	2.77	4.88	8.46	12.4	19.5	27.7
	3.0	1.69	2.42	3.91	6.23	8.63	12.8	17.0
	4.0	1.58	2.17	3.34	5.13	6.94	9.97	12.9
	6.0	1.46	1.91	2.76	3.99	5.18	7.09	8.85
	8.0	1.38	1.74	2.40	3.34	4.25	5.66	6.95
	10.0	1.33	1.63	2.19	2.97	3.72	4.90	5.98
Aluminum	0.5	2.37	4.24	9.47	21.5	38.9	80.8	141
	1.0	2.02	3.31	6.57	13.1	21.2	37.9	58.5
	2.0	1.75	2.61	4.62	8.05	11.9	18.7	26.3
	3.0	1.64	2.32	3.78	6.14	8.65	13.0	17.7
	4.0	1.53	2.08	3.22	5.01	6.88	10.1	13.4
	6.0	1.42	1.85	2.70	4.06	5.49	7.97	10.4
	8.0	1.34	1.68	2.37	3.45	4.58	6.56	8.52
	10.0	1.28	1.55	2.12	3.01	3.96	5.63	7.32
Iron	0.5	1.98	3.09	5.98	11.7	19.2	35.4	55.6
	1.0	1.87	2.89	5.39	10.2	16.2	28.3	42.7
	2.0	1.76	2.43	4.13	7.25	10.9	17.6	25.1
	3.0	1.55	2.15	3.51	5.85	8.51	13.5	19.1
	4.0	1.45	1.94	3.03	4.91	7.11	11.2	16.0
	6.0	1.34	1.72	2.58	4.14	6.02	9.89	14.7
	8.0	1.27	1.56	2.23	3.49	5.07	8.50	13.0
	10.0	1.20	1.42	1.95	2.99	4.35	7.54	12.4
Lead	0.5	1.24	1.42	1.69	2.00	2.27	2.65	(2.73)
	1.0	1.37	1.69	2.26	3.02	3.74	4.81	5.86
	2.0	1.39	1.76	2.51	3.66	4.84	6.87	9.00
	3.0	1.34	1.68	2.43	2.75	5.30	8.44	12.3
	4.0	1.27	1.56	2.25	3.61	5.44	9.80	16.3
	5.1087	1.21	1.46	2.08	3.44	5.55	11.7	23.6
	6.0	1.18	1.40	1.97	3.34	5.69	13.8	32.7
	8.0	1.14	1.30	1.74	2.89	5.07	14.1	44.6
	10.0	1.11	1.23	1.58	2.52	4.34	12.5	39.2

*μX = mass absorption coefficient (μ/ρ) x shield thickness (cm) x shield density (g/cm^2).

NOTE: For concrete, use an average of aluminum and iron; e.g., B (concrete) = [B(iron) + B(Al)]/2.

Source: "Radiological Health Handbook," U.S. DHEW, Public Health Service, 1970.

formula will express exposure in terms of milliroentgens per hour and is as follows:

$$mR/hr = \frac{5000\ CEf}{d^2}$$

where
C = activity (in microcuries)
E = energy in million electron volts (MeV)
f = the fraction of disintegration emitting E
d = the distance in cm

If the distance is to be calculated in feet, then the constant, 5000, must be replaced with the constant, 6. Further, if more than one level of photon energy is given off, each energy level (E) times its fraction of disintegration (f) must be calculated and a cumulative total of the exposure made.

This formula has two values. First, if a new source is being introduced, it is possible to calculate the exposure rate of that source at any distance. Second, in an emergency, it is possible to calculate the approximate safe distance that will not exceed the maximum permissible dose (MPD) limits. Table 6.5.7 presents data on typical radionuclides as found in the literature. This information will be used in subsequent examples and practice exercises.

Example

A 50 microcurie source of Krypton-85 is to be used in a new industrial process being introduced in your facility. What is the exposure rate of an unshielded source at 25 cm?

Solution

$$mR/hr = \frac{5000\ CEf}{d^2}$$

where
C = 50 microcuries
E = 0.514 million electron volts
f = 0.43%
d = 25 centimeters

Therefore

$$mR/hr = \frac{5000 \times (0.514 \times 0.0043) \times 50}{25^2}$$

$$= 0.885\ mR/hr \text{ at } 25\ cm$$

It is also possible to perform the same calculation using the gamma ray constant (Γ) in the formula.

$$R/hr = \Gamma A/d^2$$

Table 6.5.7

Data on radionuclides referred to in problems.

Radionuclide	Symbol	Half-Life	Major radiations, energies (MeV) and percent of disintegrations	Specific gamma ray constant, R-cm^2/hr-mCi
Calcium-45	Ca-45	165 days	β^-0.257 max; 0.077 avg	
Carbon-14	C-14	5730 yr	β^-0.156 max; 0.049 avg	
Cesium-137	Cs-137	30 yr	β^-1.174 max; (5.9%) 0.512 max; (94.1%) 0.188 avg/dis e$^-$0.624 (7.8%); 0.656 (1.7%) γ 0.662 (85%) + low energy barium X-rays	3.3
Cobalt-60	Co-60	5.3 yr	β^-1.49 max (0.12%) 0.318 max (100%) 0.087 avg/dis γ 1.332 (100%); 1.173 (100%)	13.2
Gold-198	Au-198	2.70 days	β^-0.961 max (99%) 0.28 max (1%); 0.32 avg/dis e$^-$0.33(2.9%); 0.40 (1.4%) γ 1.088 (0.2%) 0.676 (1%); 0.412 (95%)	2.35
Hydrogen-3	H-3	12.3 yr	β^-0.0185 max; 0.0057 avg	
Iodine-131	I-131	8.05 days	β^-0.807 max (0.7%) 0.606 max (90%) 0.33 max (7%); 0.18 avg/dis e$^-$0.33 (1.7%); 0.046 (3%) γ 0.723 (1.6%); 0.637 (6.9%); 0.364 (82%) 0.284 (5.8%); 0.08 (2.5%)	2.18
Iron-59	Fe-59	45.6 days	β^-1.57 max (0.3%); 0.47 max (53%) 0.27 max (46%) 0.188 avg/dis γ 1.292 (44%); 1.099 (56%); 0.192 (2.8%) 0.143 (0.8%)	6.13
Krypton-85	Kr-85	10.3 yr	β^-0.67 max; 0.246 avg γ 0.514 (0.43%)	1.29

Table 6.5.7 (Continued)

Radionuclide	Symbol	Half-Life	Major radiations, energies (MeV) and percent of disintegrations	Specific gamma ray constant R-cm^2/hr-mCi
Molybdenum-99 + Technetium-99m	Mo-99	67 hr	β^-1.23 max (80%); 0.454 max (19%) γ 0.392 avg/dis 0.78 (4.8%); 0.74 (14%) (4.8%); 0.74 (14%) 0.37 (1.5%); 0.18 (6.7%); 0.14 (90%) 0.04 (1.3%) 0.019 (12%); + low-energy Tc-99 X-rays	1.29
Phosphorus-32	P-32	14.3 days	β^-1.71 max; 0.695 avg	
Potassium-42	K-42	12.4 hr	β^-3.52 max (82%); 2.0 max (18%); 1.43 avg dis γ 0.31 (0.2%); 1.52 (18%)	1.50
Sodium-24	Na-24	15.0 hr	β^-1.389 max; 0.55 avg γ 2.754 (100%); 1.369 (100%)	18.8 (11.7 from 2.75 MeV and 7.1 from 1.37 MeV)
Sulfur-35	S-35	87.9 days	β^-0.167 max; 0.049 avg	
Technetium-99m	Tc-99m	6.0 hr	e$^-$0.138 (1.1%); 0.199 (8.8%); 0.0017 (98.6%); 0.017 avg/dis γ 0.140 (90%) + low-energy X-rays	0.70
Xenon-133	Xe-133	5.3 days	β^-0.346 max (99.1%) 0.266 max (0.9%); 0.10 avg e$^-$0.80 (2.4%); 0.075 (8.7%); 0.045 (52%) γ 0.081 (37%); Cs X-rays	

Sources: Martin and Blichert-Toft, 1970; Dillman, 1969; Hine and Brownell, 1956 (for values of specific gamma-ray constants)

where

Γ = gamma ray constant (R-cm^2/hr-mCi)
A = activity (mCi)
d = distance (cm)

Using the tools of time, distance, and shielding, the exposure to employees can be minimized. By varying any of these factors, an increase or decrease in exposure can occur. In any instance, the exposure level can be minimized so that it is below established standards. The sample problems presented at the end of this chapter provide an opportunity to apply the concepts just presented on the shielding requirements of gamma radiation.

X-Radiation Shielding

When evaluating the potential radiation hazard produced by an X-ray machine, the following three components must be considered. The first component is the useful beam itself. This is the beam and its energy directed at a specified target. The second component is scattered radiation. This is the incident radiation that is reflecting off the target, the walls, etc. Finally, leakage radiation must be considered. This is the radiation that penetrates the tube housing and is not part of the useful beam. The degree of shielding and protection required for an X-ray installation varies. X-ray shielding is based on the same principles as gamma shielding. However, direct calculation of half value layers for X-ray shielding are difficult. Factors such as the complexity of the X-ray spectrum, dependence upon the width of the beam and diffraction patterns, and factors contributing to the beam's scattering make HVL calculations difficult. Generally, when developing shielding patterns for the X-ray area, it is better to rely on the calculated values in the literature that are based on experiments that have a similar design. For example, Table 6.5.8 illustrates the half value layers for diagnostic X-rays for two shielding materials from 50 to 100,000 volts. The shielding is designed to limit the maximum exposure to 0.1 roentgens per week at specified dose point.

Table 6.5.8

Half-value layers for diagnostic X-rays.

Tube Voltage	Lead (mm)	Concrete[a]
50,000	0.05	0.43
70,000	0.15	0.84
100,000	0.24	1.50
125,000	0.27	2.00

(a) The half-value layer will vary in different kinds of concrete and is given for illustrative purposes only.

Source: NCRP 1970, Report 34.

When evaluating the potential X-ray exposure that may be produced by a given installation, the following factors must be considered:

1. Work load (W)--The degree of use of the machine. This is usually expressed in terms of milliamp-minutes per week (mamp-min/wk).

2. Use factor (U)--Fraction of work load during which radiation under consideration is pointed in the direction of interest; e.g., toward personnel.

3. Occupancy factor (T)--Factor of occupancy of areas of interest.

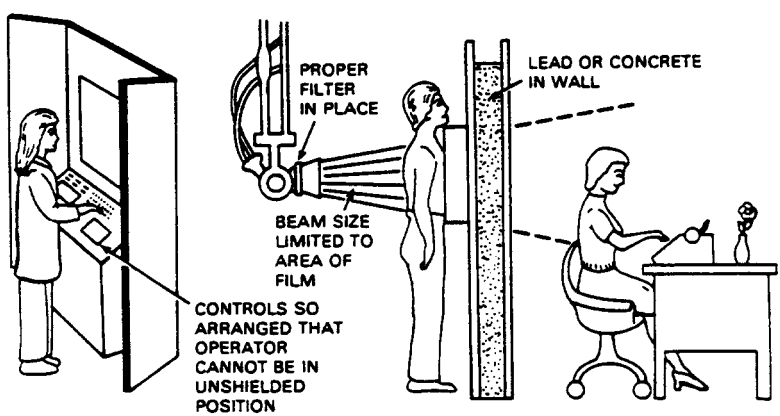
4. Output (P)--The output of the direct beam.

5. The distance from the beam source to the employees of interest.

Figure 6.5.6 illustrates a typical X-ray installation.

Figure 6.5.6

Medical X-ray protection.



If the output, work load, use factor, and occupancy factor are known, shielding requirements can be determined. First, it is necessary to calculate the output from the tube (TO). This can be done using tables found in the literature such as illustrated in Table 6.5.9. Then, it is necessary to calculate the product of W x U x T. Then, the attenuation can be calculated by determining the product of the output specified for the distance from the target to the point of interest and the three factors, W, U, T. This product is divided into the permissible weekly dose. The shield thickness is then calculated in the usual way from the number of half value layers required to produce the desired attenuation. For convenience, as previously mentioned, the protection handbooks contain tables for determining shielding thickness at installations operated at a given kilovolt peak.

Table 6.5.9

Average radiologic output*

Target Distance		Tube Potential (ρ)						
		50 Kvp	60 Kvp	70 Kvp	80 Kvp	90 Kvp	100 Kvp	125 Kvp
inches	cm	R/100 mamp-sec						
12	30	1.8	2.8	4.0	5.8	8.0	9.8	15.2
18	46	0.8	1.3	1.8	2.5	3.4	4.2	6.7
24	61	0.4	0.7	1.1	1.4	1.9	2.3	3.8
36	91	0.2	0.3	0.5	0.6	0.9	2.1	1.7
54	137	0.1	0.1	0.2	0.3	0.4	0.5	0.7
72	183	0.1	0.1	0.1	0.2	0.2	0.3	0.4

* Measured in air with total filtration equivalent to 2.5 mm Al.
Source: NBS Handbook 76, 1961, Appendix A, Table 7.

Example

Determine the thickness of lead required on the wall of a radiographic installation with personnel working 12 feet from the source on the other side of the wall with the following conditions:
Kvp = 70
W = 200 milliamp minutes per week
U = 1
T = 0.25

Solution

A. Calculate tube output (TO). (From Table 6.5.9)
at 1 ft and 70 Kvp
TO = 4.0 R/100 mamp-sec
at 12 ft

$$TO = \frac{4.0 \text{ R/100 mamp-sec}}{(12)^2} = 0.028 \text{ R/100 mamp-sec}$$

B. Calculate W x U x T
W = 200 mamp-min/wk
U = 1
T = 0.25
W x U x T = 200 mamp-min/wk x 1 x 0.25
= 50 mamp-min/wk

C. Calculate attenuation

$$\text{attenuation} = 0.1 \text{ R/wk} \times \frac{1}{50 \text{ mamp-min/wk}} \times \frac{100 \text{ mamp-sec}}{0.028 \text{ R}} \times \frac{1}{60 \text{ sec/min}}$$

attenuation = 0.12

D. Calculate HVL

$$(0.5)^n = 0.12$$

$$n = \frac{\log (0.12)}{\log (0.5)} = \frac{-0.9208}{-0.3010} = 3.06$$

From Table 6.5.8, it is found that 3.06 x 0.15 = 0.459 mm lead is required to shield this source.

The calculation for shielding requirements for incident radiation may also be calculated in the same manner. As previously mentioned, this procedure is somewhat simplified in the literature, but the same information is required to manipulate the data and calculate the shielding requirements.

The shielding requirements for beta radiation are somewhat less than for gamma and X-radiation because of the lesser penetrating capability of beta radiation. The range of the beta particle is a function of energy level of the beta particle and the composition of the absorbing material. The "range" of a beta particle refers to the thickness of material through which no beta particle emitted from a source can penetrate. Any shielding greater than the range specified for a given beta particle will prevent any emission of beta particles through the shield. In the literature, the range for the various energy levels of beta particles is given in terms of unit density material. Table 6.5.10 gives some properties of commonly used beta emitters including the unit density material. The range for the unit density material is based upon the assumption that the density of this absorbing material is one gram per cubic centimeter.

If an absorbing material with a density other than 1.0 gram per cubic centimeter is used, then the range can be calculated by dividing the unit density by the density of the shielding material. For example, if the minimal thickness of the wall of a glass test tube required to stop all beta particles from a Phosphorus-32 source is to be calculated, it is necessary to divide 0.8 cm (the unit density for P-32) by 2.3 grams/cm^3 (the density of glass) to get a value of 0.35 cm of glass as the range for Phosphorus-32 beta particles.

When considering the type of shielding material used for beta particle shielding, special consideration must be given to the atomic weight of the shielding material. Beta particles should be shielded by materials that are light or have small atomic weights; e.g., aluminum, water, glass. This is necessary because the beta particles convert to the more penetrating X-radiation in shielding materials like lead that have large atomic numbers. Because this production is more likely with heavier compounds, beta particle shielding should always be done with light atomic weight materials.

As with beta particles, alpha particles, because of their high atomic weight and charge, have a limited penetrating capability and range. This is true to the extent that alpha radiation cannot penetrate the outer layer of dead skin of the body. However, the danger with alpha radiation is the potential of the alpha emitter entering the body. Therefore, containment more

than shielding is required when working with alpha emitters. Figure 6.5.7 illustrates a typical installation used for the containment of alpha particles; e.g., the glove box. The glove box will be discussed in greater detail in this chapter.

The final radiation form that has been discussed, neutrons, can be shielded following the same concepts of half value layer and attenuation coefficients as described with gamma photon shielding. The most efficient absorption materials used for neutron shielding are high (> 20%) in hydrogen content. Figure 6.5.8 illustrates the attenuation of hydrogen per energy level of neutrons.

To calculate the attenuation for other absorbing medium used as shielding, it is first necessary to calculate the hydrogen density in that medium and then multiply that by the attenuation coefficient found on the table. To calculate the half value layer, it is then necessary to divide the attenuation coefficient calculated into the constant, 0.693, so that

$$HVL = \frac{0.693}{\text{attenuation coefficient}}$$

Table 6.5.10

Properties of some commonly used beta emitters.

Property	H-3	C-14	S-35	Ca-45	P-32	Sr-90
Half-Life	12.3 yr	5730 yr	88 d	165 d	14.3 d	28.1 yr
Maximum beta energy (MeV)	0.018	0.154	0.167	0.254	1.71	2.24*
Average beta energy (MeV)	0.006	0.050	0.049	0.077	0.70	0.93
Range in air (ft)	0.02	1	1	2	20	29
Range in unit density material (cm)	0.00052	0.029	0.032	0.06	0.8	1.1
Half value layer, unit density absorber (cm)	--	0.0022	0.0025	0.0048	0.10	0.14
Dose from 100 beta particles/cm^2-sec (mrad/hr)	--	64	60	43	12	11
Fraction transmitted through dead layer of skin (0.007 cm)	--	0.11	0.16	0.37	0.95	0.97
Dose rate to basal cells of epidermis from 1 $\mu Ci/cm^2$ (mrad/hr)	--	2600	3600	5900	4300	3900

* From the Y-90 decay product.

Source: Shapiro, Jacob. Radiation Protection--A Guide for Scientists and Physicians, Harvard University Press, 1974.

Figure 6.5.7

Glove box.

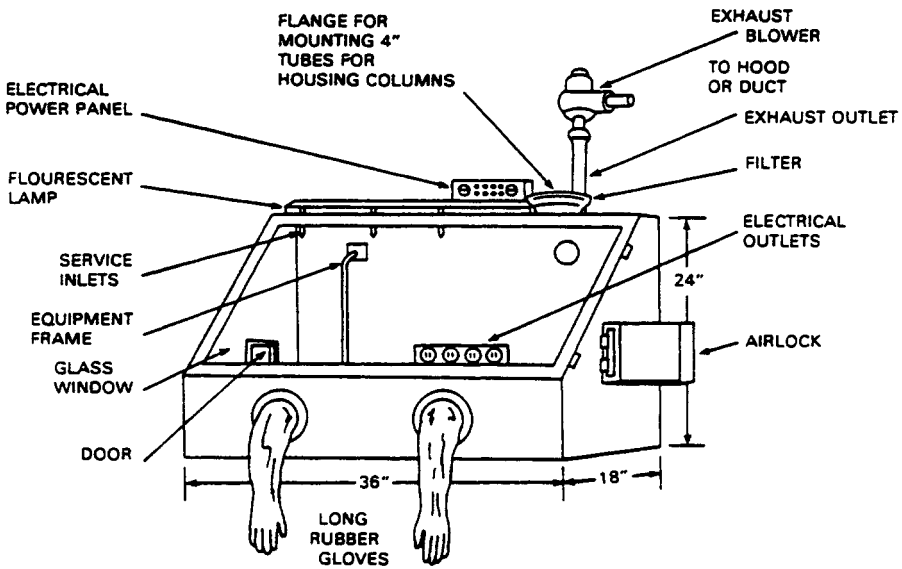
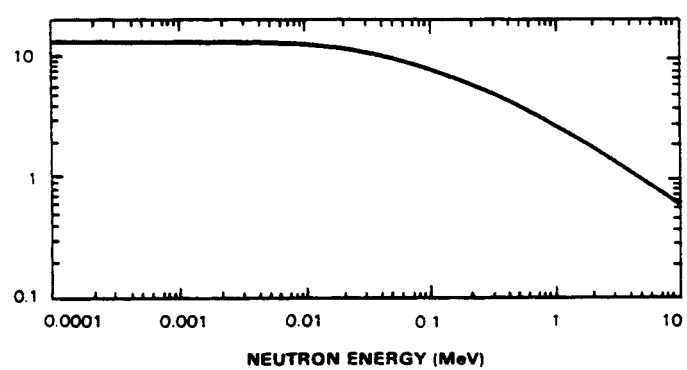


Figure 6.5.8

Hydrogen attenuation (cm^{-1}) vs. neutron energy.



The following is an example of the calculation of the attenuation due to hydrogen in a water shield 150 cm thick for 8 MeV neutrons.

Water is 11% hydrogen by weight. Therefore, the density of hydrogen in the medium, water, is 0.11 grams/cc. From Figure 6.5.8, for neutrons with 8.0 MeV energy level, the attenuation coefficient equals 0.68 cm^{-1}. Therefore, for water the attenuation coefficient would be 0.68 x 0.11 or0.075 cm^{-1}. The half-value layer would then be 0.693 divided by 0.075, or 9.24 centimeters. The number of half-value layers would be equal to 150 divided by 9.24, or 16.2. The attenuation caused by 16.2 half-value layers would be equal to $(0.5)^{16.2}$, or 1.3×10^{-5}.

Application of the previous formula can be used to calculate the effect of neutron shielding. However, neutron activity leads to the emission of gamma radiation. Therefore, neutron shielding also requires gamma shielding. Because of the sophistication of this problem, it is beyond the scope of this text. Any situations that require neutron shielding should be investigated further in the literature.

Survey and Monitoring Procedures for Radiation Hazards

The purpose of surveying and monitoring areas with potential radiation hazards is:

1. To indicate the general level of radiation in the working environment.
2. To monitor changes in the level of radiation.

An area surveyed would include an initial examination of the facility, laboratory operating procedures, personnel habits, types of radiation sources being used, methods used in handling the sources, and radiation levels that are observed. Special emphasis should be placed on identifying and quantifying those factors that determine the exposure rate for the personnel. These things include the average time of exposure during the operating procedure, average distance of the employee from the potential source, and areas of the body that may come in closer contact to the source than others, e.g., the hand. Area surveys are generally more interested in analyzing potential radiation hazards and identifying existing hazards without actually quantifying the hazard.

These surveys should be scheduled on a routine basis. Shields should be checked for cracks, excessive surface and airborne contamination should be evaluated, and factors such as improper disposal of waste should also be considered. Special emphasis should be placed on unlikely or remote areas that typically are not included in any type of daily monitoring procedure.

Routine measurements should be made at intervals during the performance of work. Measurements should be taken for all steps of the procedure and should include data such as the time of the procedure and exposure levels; distance from the source, including the mean body distance and specific areas of the body that are closer to the source, e.g., hands; the variations of exposure

throughout the procedure to the employee, e.g., does the employee turn his back to the source during the procedure? From this information, a relative exposure rate for personnel can be determined and adjustments made accordingly. Care should be taken when taking these measurements to ensure that all potential exposed areas are analyzed.

Efforts should be made, if appropriate, to analyze the contamination that may occur during any given procedure. Specifically, three areas are of concern. First, any radioactive contamination that may settle on any surface in the laboratory in the form of dust or particulate matter. This may be evaluated by taking a smear or wipe test. Its purpose is to determine the amount of loose radioactive material that may potentially become airborne or be transferred to personnel and carried outside the radiation facility. If a specific surface area, for example 100 cm^2, is wiped off with a clean cloth or adhesive tape each time a smear test is performed, then the results may be quantified. Smear samples, once taken, should be removed to areas of low background radiation and a reading taken. If surface contamination is found beyond established limits, corrective action should be taken.

The second form of contamination that should be monitored is that of air contamination. Air samples may be collected by standard procedure, such as using filters, electrostatic precipitators, impingers, or impactors. Care must be taken to ensure that all particles of appropriate size are collected. Samples of 10 m^3 are usually adequate. Direct counting from the surfaces where air samples are collected is appropriate. Care must be taken when counting alpha particles because the alpha particles may be absorbed by the sample-collecting filter. In this instance, a correction factor must be determined. Assuming that the radionuclide is known, the concentration of radioactive contaminant in the air can be calculated in terms of microcuries per cubic centimeter (μCi/cc). This value is then compared to the maximum permissible concentration (MPC) for unrestricted areas. These values may be found in Handbook 69. (See References.) Once the concentration has been calculated, the quantity of nuclide taken in by personnel can be roughly estimated by determining the exposure time and then determining the product of the exposure time, concentration, and the conversion factor of 10^7 cc per eight hours. This value can be compared to the maximum body burden, also found in Handbook 69. If the nuclide source is not known, arbitrary limits can be established such that gamma and beta sources should not exceed 10^{-9} μCi/cc, and alpha sources should not exceed 10^{-12} μCi/cc. In any instance where air contamination is a possibility, respirators should be used until radiation levels have been adequately determined. In some instances, naturally occurring radon and thoron may interfere with readings; specifically when counting alpha particles. In this instance, adjustments must be made for the interference.

Water sample analysis is similar to air analysis. The sample should be 100 to 500 ml. In analyzing the sample, the water is evaporated and a reading is taken. The concentration in microcuries per cubic centimeter is then calculated and the results compared to the maximum permissible concentration (MPC) values, also found in Handbook 69. If the nuclide is not known, qualitative tests can be performed to determine the source. In either the air or the water sample analysis, the minimum level for the critical organ should be used as a basis for comparison.

Personnel Monitoring

Personnel monitoring is the most direct method of calculating personnel exposure. As previously discussed, the common equipment used for personnel monitoring includes film badges, pocket ion chambers, and pocket dosimeters. Personnel monitoring should be used in any situation where it is possible that 25% of the MPD value for a 13-week period may be exceeded. Because of the potential liability of the instruments, the personnel devices should always be used in pairs.

The location of the personnel monitor is critical. If the whole-body radiation dose is of interest, the monitor should be worn between the waist and the neck, somewhere on the chest. It should be unshielded; that is, the monitor should not be covered by several layers of clothing. To maximize the value of the monitor, it should be at a point of maximum exposure. If, for example, the process requires the employee to handle the radioactive source with leaded gloves on, the monitor should be placed on the hands rather than somewhere on the body. Although the MPD value is greater for the hands and a body exposure may give a relative indication of exposure to the hands, the most accurate reading is necessary. The configuration of the employee's body to the radiation source must also be considered. If, for example, the person works 50 percent of the time with his left side facing the radiation source and 50 percent of the time facing the source directly, then the monitor should be placed such that readings are being taken at the points of maximum exposure and at no time does the worker's body shield the monitor from the radiation source.

Special effort must be made to ensure that the monitors selected match the type and energy level of the radiation being monitored. The use of monitors and dosimeters that are not correctly matched will give erroneous readings. Records on each employee should be maintained concerning cumulative doses received.

In some instances, it may be of value to receive a continuous recording of dose rate at a fixed location. In these instances, fixed monitors are used. These monitors may be equipped with visible or audible alarms that warn of increased radiation levels. If contamination is possible, a fixed monitor may be placed in the doorway of the radiation facility so that personnel are monitored for contamination of clothing before leaving the facility. Special monitors have been developed that survey the hands and shoes, including the soles. This type of monitoring capability is of value when large numbers of personnel are involved at the change of shifts. It limits the possibility of radioactive contamination outside the designated facility.

Controls can be designed into the system that minimize personnel exposure. These include warning systems, such as previously discussed, for high exposure levels. Also, it is possible to develop interlock systems so that radiation-producing equipment (X-rays) cannot be operated unless shielding is in place, or shielding cannot be moved until personnel are a designated distance from the source.

Facilities

When examining the design of the facility to be used for a radioactive procedure, steps must be taken to maximize containment of the radiation and to allow for the ease of cleanup in the event of contamination within the facility. Certain design factors can maximize this.

All surfaces involved in the radiation area should be smooth and nonporous; e.g., shelves, floors, sinks. It is unacceptable to have surfaces--such as uncoated wood, concrete, or soapstone--that would provide a porous surface for the collection of contaminants. Such surfaces as tile, polished stainless steel, and plate glass are acceptable. Although paints, varnishes, and lacquers do provide a somewhat nonporous surface, they are not recommended for use in radiation areas.

Any area that could be a dust collector should be eliminated. This includes things such as suspended lighting, suspended pipes, and roof trusses. Any shelving or storage areas should be enclosed by doors. Cove corners between the wall and floor will facilitate cleanup and reduce possible gathering of contaminated particles.

Any special piping or drainage systems that are necessary for the procedure and are involved in the radiation area should be plainly labeled, especially if used for radioactive waste.

The design of the facility and shielding should prevent any type of radiation leakage from the facility. For example, Figure 6.5.9 illustrates special precautions that must be taken in the construction of joints to prevent the leakage of incident radiation. Figure 6.5.10 illustrates a typical doorway maze that is used to minimize leakage.

Within the facility, special equipment is often required. For example, long-handled tools such as tongs and forceps are often used to increase the distance between personnel and the source and decrease the exposure rate (inverse-square law). In those instances when the radiation source is high-activity, it may be necessary to use remote control devices with protective lead glass windows such as displayed in Figure 6.5.5.

With respect to the facility design, there are two areas that should be given special consideration. First, when working with unsealed radioisotopes, hood and exhaust systems are needed to minimize airborne contamination. The airflow in the hood system should result in a minimum average face velocity of 100 feet per minute (fpm). If highly toxic radioisotopes are being used, then the flow rate should be adjusted to achieve a average face velocity of approximately 125 to 150 fpm. The hood should have its own exhaust system with appropriate filtering and decontamination systems. The air contamination should be monitored to ensure that it does not exceed established levels. (Note: The calculation of air contamination will be discussed later.) Airflow should also be monitored for cross drafts and leaks in the system. The development of the hood and exhaust system should follow the principles discussed in the Ventilation section.

Figure 6.5.9

Joint construction.

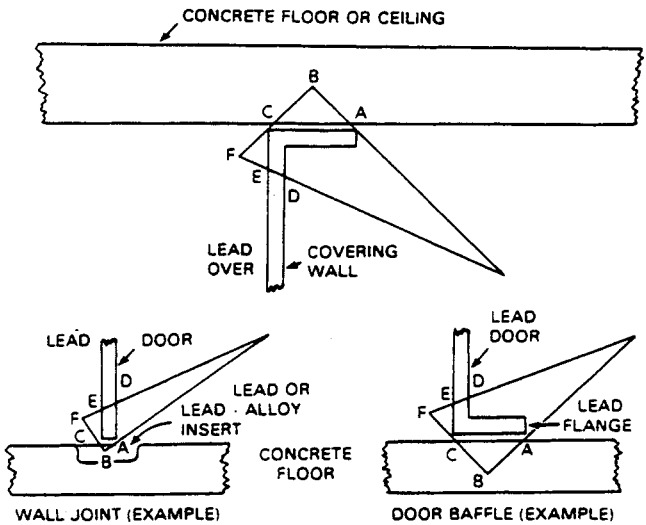
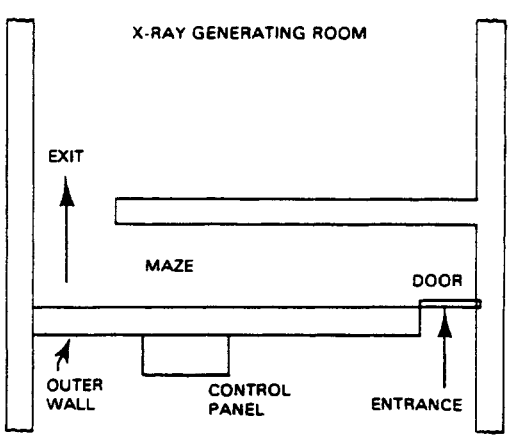


Figure 6.5.10

Entrance maze.



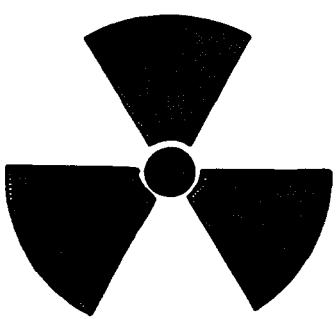
When working with alpha and low-energy beta radiation, glove boxes, as shown in Figure 6.5.7, offer sufficient protection. This type of system prevents air contamination while allowing the employee to work closely with the radioisotope. As seen in the figure, hermetically sealed ports go into the box and allow for manipulation of the box contents using the glove. Air locks are also provided to allow for insertion and removal of samples without air contamination. For high-energy beta rays and gamma radiation, it is necessary to replace the gloves with some type of remote mechanical manipulator. Frequently, the glove boxes have exhaust ports and filters with exhaust volumes in the area of 20 to 30 cubic feet per minute. The personnel should be positioned so that any explosion would not be directed at them through exhaust and inlet ports.

Posting the Area

Warning signs are essential in radiation areas since individuals might otherwise be unaware of the presence of the radiation field. Areas need to be posted only if the radioactivity is to be present in that area for longer than eight hours. Each symbol or label should bear the three-blade radioactive caution symbol, either magenta or purple on a yellow background, as shown in Figure 6.5.11.

Figure 6.5.11

Radiation symbol.



The labels and signs that accompany the radiation symbol are dependent upon the potential exposure in that given area. The following types of signs are required in areas where significant levels of radiation or radioactivity are present:

1. "CAUTION RADIATION AREA"--This sign is used in areas accessible to personnel in which a major portion of the body could receive in any one hour a dose of 5 millirems, or in any 5 consecutive days, a dose in excess of 100 millirems.

2. "CAUTION RADIOACTIVE MATERIAL"--This sign is required in areas or rooms in which radioactive material is stored or used in an amount exceeding the quantities listed in Table 6.5.1.

3. "CAUTION RADIOACTIVE MATERIAL" (Label)--A durable, clearly visible label is required on any container that is transported, stored, or used for a quantity of any material greater than the quantity specified in Table 6.5.1. When containers are used for storage, the labels must state also the quantities and kinds of radioactive materials in the containers and the date of measurement of the quantities.

4. "AIRBORNE RADIOACTIVITY AREA"--This sign is required if airborne radioactive activity exceeds at any time concentrations in excess of the maximum permissible for 40 hours of occupational exposure, or if the average over a number of areas in any week during which individuals are in the area exceeds 25 percent of the maximum permissible concentration. (Table 6.5.1)

5. "HIGH RADIATION AREA"--This sign is required if the radiation dose to a major portion of the body to a person in the area could be in excess of 100 millirems in any one hour. These areas also require the audible or visible alarm signals that were discussed previously.

As stated, because radioactivity cannot be sensed by the human body, it is essential that signs be present in areas where radioactivity may exist. On the other hand, the signs should not be used when they are not needed.

In many instances, the handling procedures for radioisotopes or the operation of radiation-producing instruments have been specifically defined. In those instances, it may be of value to post the operating/handling procedures as illustrated in Figure 6.5.12.

Trays and Handling Tools. When working with radioisotopes, any procedure that may result in the contamination of a table top should be performed in a tray. This would negate the problem of having to replace counter tops rather than the contaminated trays. Also, even small amounts of radionuclides should not be handled directly, but tweezers and/or tongs should be used whenever possible to minimize exposure. This will reduce the potential of contamination on the hands and gloves and also decrease the exposure rate through the inverse-square law of distance.

Storage and Disposal of Radionuclides

Radionuclides should be stored in designated areas, protected against fire, explosion, or flooding. They should be stored in suitable containers that provide adequate shielding. In any type of storage situation, the radiation level should not exceed 5 millirems per hour at one foot in the storage area.

Figure 6.5.12

How to handle radioisotope shipments.

An Outline of Recommended Procedures to be Followed When Receiving Radioactive Shipments

Open and inspect packages immediately upon receipt.
Radioactive solutions inadvertently stored upside down may gradually leak and cause contamination problems; furthermore, vendors often will not accept claims for shipments not inspected within 15 days after delivery.

Monitor package for radiation field.
It is suggested that plastic gloves be worn while processing the received package.

To Process Soft Beta, Hard Beta, and Gamma Emitters

1. Wipe and test package for removable contamination.
2. Note radiation units stated on package, verify and record in receipt log. (Hard beta and gamma only.)
3. Place package in vented hood.
4. Open outer package and remove packing slip. Open inner package and verify that the contents agree in name and quantity with the packing slip.
5. Measure radiation field of unshielded container--if necessary, place container behind shielding to reduce field to allowable limits and proceed with remote handling devices. (Hard beta and gamma only.)
6. Check for possible breakage of seals or containers, loss of liquid, or change in color of absorbing materials.
7. Wipe test inner contents and document any pertinent findings on packing slip. Note: The liner, shield, and isotope container may have surface contamination; they should be discarded in hot waste disposal containers.
8. Record type of activity, quantity present, and location of delivery in receiving log.
9. Deliver processed package to proper laboratory. If delivery is delayed, notify recipient of its arrival and clearance.
10. If material has been packaged in dry ice, refrigerate or deliver immediately to ultimate user.
11. If contamination, leakage, or shortages are observed, notify the vendor's Customer Service Department immediately.

When concerned with disposal of radioactive wastes, there are four possible alternatives: release into the atmosphere, release into water, burial, or a contract arrangement with a commercially licensed radiation disposal firm. Each of these activities is controlled by the Nuclear Regulatory Commission.

The release of radioactivity in the air is limited by the Nuclear Regulatory Commission. The concentration released through exhaust system should not exceed the maximum permissible concentration (MPC) at the point of discharge. The exact concentration can be calculated, including the release of different nuclides at once.

Example

A radio chemist released 1 curie of tritium (tritiated water) through a hood while performing a synthesis. The hood face velocity was 100 feet per minute with a 1 foot by 4.5 foot hood opening. If the concentration is averaged over one week, was the MPC value exceeded?

Solution

Assuming 40 hours per week, the flow rate for the hood can be calculated to be 0.612×10^{10} cc/day or 3.06×10^{10} cc/week. From Table 6.5.11, the maximum permissible concentration for tritiated water (Hydrogen-3) is equal to 2.0×10^{-13} curies/cc. The radioactive release from the hood is equal to 1 Ci/week ÷ 3.06×10^{10} cc/week = 3.26×10^{-11} Ci/cc. Therefore, the MPC has been exceeded.

Limited amounts of liquid radioactive waste can be deposited in unrestricted water or in sewage systems. Again. levels have been established by the NRC and must be met. Unlike the calculation of maximum level for air contamination, when calculating liquid disposal, both daily and monthly MPC values must be met. Further, disposal may not exceed 1.0 curies per year. An example of this is as follows:

Example

Determine how much Iodine-125 and Phosphorus-32 can be discharged into the sewerage system if the water flow (based on water bills) is 1.2×10^7 cubic feet per year.

Solution

From Table 6.5.11, the MPC value for Iodine-125 is 40×10^{-12} Ci/cc/wk, and the MPC value for Phosphorus-32 is 5.0×10^{-10} Ci/cc/day. The average daily water flow can be calculated based on the water bill to be equal to 9.31×10^8 cc/day (assume 365 day operation). Therefore, the daily limits for Iodine-125 and Phosphorus-32 would be equal to the product of the MPC value times the daily water flow, or 37.24 millicuries and 466 millicuries respectively. Therefore, these daily limits would be the maximum

amount of radioactive waste that could be discharged into the sewerage system. However, this daily disposal rate could only continue until a maximum of 1 curie/yr limit for the disposal of gross activity has been reached.

When disposing of solid waste, the alternative of incineration, burial, or the commercial disposal firm are available. Incineration is a good bulk-reducing method. Again, this method is regulated by the NRC. The rule of thumb to follow is that the concentration released to the unrestricted areas should not exceed limits specified for continuous exposure. When calculating the release, the concentration may be averaged over a maximum of one year. The procedure for calculation is similar to that for air contamination, and to complete the calculation, the type of radionuclide, the quantity of radionuclide, and the airflow of the incinerator must be known. An important fact is that any ash left over after incineration must also be treated as radioactive waste.

Table 6.5.11

Maximum permissible concentrations of radionuclides in air and water.

	Unrestricted Areas		Restricted Area (40 hr/wk)	
Radionuclide	Water (pCi/cc)	Air (pCi/cc)	Water (pCi/cc)	Air (pCi/cc)
C-14	80	0.1	20,000	4
H-3	3,000	.2	100,000	5
S-35	60	.009	2,000	0.3
I-131	0.3	.0001	60	0.009
I-125	0.2	.00008	40	0.005
P-32	20	.002	500	0.07
Ca-45	9	.001	300	0.03
Na-24	30	.005	800	0.1
K-42	20	.004	600	0.1
Cr-51	200	.08	50,000	2
Kr-85	--	.3	--	10
Xe-133	--	.3	--	10
Br-82	40	.006	1,000	0.2
Cl-36	60	.008	2,000	0.02

Note: The maximum permissible concentration depends upon several factors, including the degree of solubility of the contaminant. The lowest concentrations specified in the regulations have been listed.

Source: U.S. Code of Federal Regulations, Title 10, Part 20, as of December 10, 1969.

The NRC rules for burial are fairly simple. Each organization is permitted twelve burials per year. Burials must be at least six feet apart and a minimum of four feet deep. The total quantity of radionuclide buried at any location may not exceed 1000 times the level established in Table 6.5.1. If several nuclides are buried in a single burial, the sum of the radionuclide fractions cannot exceed one. For example, a user of radionuclides plans to dispose of 2 millicuries of Iron-59, 10 millicuries of Chromium-51, and 20 millicuries of Iodine-125 by burial. Is this possible?

Solution

He would be permitted to bury single (1000 times value in Table 6.5.1, third column):

Iron-59 = 10 millicuries
Chromium-51 = 1000 millicuries
Iodine-125 = 1 millicurie

Immediately it can be seen that the amount of Iodine-125 which is to be buried exceeds the limit. However, the Iron-59 requires only 20 percent of the allotted limit, and the Chromium-51 required only one percent of the allotted limit. Therefore, being sure that the sum of the nuclide fractions does not exceed 1, 0.79 of the Iodine limit, or 0.79 millicuries, could be buried in the same burial. However, the remaining 19.21 millicuries of Iodine would have to be buried in 19 separate burials. Because this exceeds the limit, it would seem more advisable to have the Iodine disposed of by a commercial firm.

In many instances, the guidelines for storage and disposal of wastes established by the NRC vary somewhat from state regulations. Therefore, state health departments must also be contacted when working with radioisotopes in production lines.

Personnel

Personnel have a responsibility to minimize their exposure to ionizing radiation. Although maximum exposure levels (MPD) have been established, every effort should be made to minimize exposure.

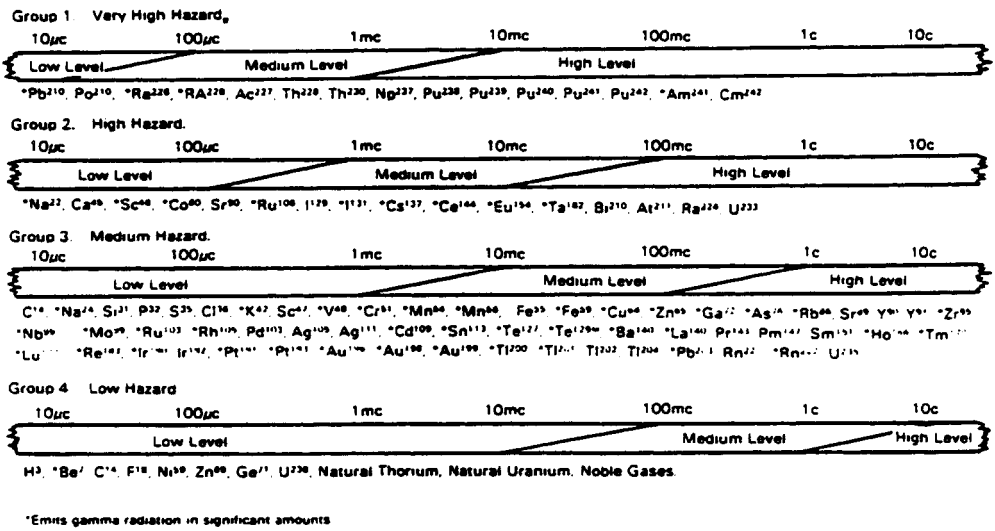
The Nuclear Regulatory Commission has established training requirements for those personnel directly involved with the use of radioisotopes or radiation-producing machinery. Each person potentially exposed to radioactivity should receive training in the potential dangers of radiation, operational procedures that minimize exposure, and procedures in case of accidents.

The protective clothing that is required of personnel is dependent on the level of radio toxicity of the nuclide. Specifically, the clothing should be such that it can be easily laundered or disposed of should contamination occur. The degree of protection that is required of the clothing is a function of the quantity and type of radioactivity, the nature of the operation, and the design of the laboratory. Sealed containers do not usually require protective clothing for use. In general, protective clothing of any

kind is not required if the MPD value for the most critical organ (as defined by Handbook 69) is not and will not be exceeded. The amount of protective clothing, as previously mentioned, is dependent upon the radio toxicity. Figure 6.5.13 generally groups the radioisotopes according to toxicity.

Figure 6.5.13

Hazard from absorption into the body.



For those personnel working with low level radioisotopes, laboratory coats or coveralls are recommended. Simple cloth or plastic bags on the shoes to reduce the potential contamination of the shoes is also recommended. Rubber or plastic gloves should be worn, and handling of the radioisotope without gloves should be avoided. Any isotope that is classified in the medium level requires the personnel to wear coveralls, caps, gloves, and shoe covers. Every effort should be made to minimize contamination of personal clothing. Should the individual be exposed to high level radioisotopes, a multi-layer of coveralls, shoes, etc., would be required. In most instances, high level radioisotopes should not be encountered during normal operation.

If the recommended MPC level is exceeded, it is required that some type of respiratory protection be used. The type of respirator employed may vary from a simple respirator, which merely filters out particles, to some type of self-contained breathing apparatus, which is a closed ventilation system for the employee. Typical respirators are illustrated in other chapters. One limitation of respiratory equipment must be mentioned; that is, in all instances the mask will leak approximately 1 to 2 percent. Growth of a beard or mustache may further affect the mask fit and leakage. Therefore, this potential leakage must be taken into consideration when the personnel are equipped with respiratory protection.

Shielding garments are somewhat different from protective clothing in that they not only serve to prevent and reduce contamination, but also they act as an absorbing or shielding material. They should be used for close contact work with radiation of low penetrating power. For example, leather or rubber gloves are often effective against most beta radiation. Fabrics loaded with high atomic material are used for shielding against X-radiation. Shielding garments should not be used solely for high-level beta or gamma radiation because the shielding cannot be designed to stop radiation of high penetrating power. Personal hygiene is also essential in minimizing exposure. Eating, smoking, food storage, pipetting by mouth should be prohibited in radioactive areas. Any work with unsealed sources requires a "wash up" before eating, smoking, or leaving the work area. Written procedures for personnel should include basic hygiene practices as mentioned.

Personnel monitoring has been discussed previously. As a review, any personnel who potentially may be in an area of radiation level that is 25 percent of the maximum permissible dose should carry personnel monitoring devices; e.g., film badges, dosimeters. As mentioned, these devices should be used in pairs at a minimum to ensure adequate coverage of the personnel.

Because of the physical hazards of ionizing radiation, medical supervision is essential. Each potential employee should have a pre-employment examination to identify his or her general physical condition. Efforts should be made at the pre-employment examination to quantify, if possible, previous radiation exposures including those involving medical examination; e.g., dental X-rays. Efforts should also be made to identify possible problems that may be magnified or potentially dangerous to the employee in a radioactive environment. Such problems might include dermatological diseases, impairment of pulmonary ventilation, or even cataracts. The decision of employment and placement should consider the above-mentioned factors. Once the employee has been hired and selected to work in a radioactive area, periodic medical examinations should be performed. They should occur at appropriate intervals based on the general health of the employee and the nature of work. These examinations should provide insight to any medical changes. However, the periodic examinations should not be used as a reliable method for monitoring radiation hazard. The examination should include a review of the occupational hazard records and the assessment of exposure doses. Medical advice should be followed with respect to continued radiation exposure for the employee.

In the event of a radiation accident, or even if the MPD value has been exceeded only slightly, it is important that followup examination continue. This is true even if the examinations must occur after the employee has resigned from the organization. This type of followup examination will provide extended coverage for the worker, along with adding information on the effect of radiation on the general population. Records should be established and maintained for each employee working in some type of radioactive environment. These records should include medical and radiation exposure history. Job assignments and hazards involved should also be recorded.

Radiation Accidents

Maximum efforts should be made to minimize accidents. For example, periodic review of operational procedures for potential hazards should be implemented. Further, equipment used in the handling and operation dealing with radioactive sources should also be checked periodically. For nonroutine or high level operations, a trial run of the operation should be performed. This trial run will evaluate the adequacy of the procedure and determine the exposure time to the personnel.

In the event of an accident, loose contamination should be minimized whenever possible. All spills should be cleaned up promptly. Cleaning tools should not be removed from the radiation area without being decontaminated. Any level of contamination is difficult to determine, and the impact of the contamination is also very complex.

When monitoring for contamination, a G-M counter is used for beta and gamma radiation, and a proportional counter is used when alpha contamination is suspected. The wipe or smear test is often performed when surface contamination is suspected.

When employees' hands, body surfaces, clothing, or shoes become contaminated, loose contamination should be removed as soon as possible. Care must be taken to minimize the spreading of the contamination. Initially, washing with mild soap or detergent is a good step. This may be followed up with a mild abrasive soap, complexing solution, or mild organic acid, whichever is appropriate. When the hands are involved in some type of contamination, clipping the fingernails may reduce contamination. In any instance, medical personnel should be notified and the employee examined. Because of potential medical treatment being based on the level of dose received and contamination involved, a relatively accurate determination of the level of exposure is necessary. With respect to specific procedures for decontamination, the following section provides a general overview for radiation accidents.

Emergency Instructions in the Event of Release of Radioactivity and Contamination of Personnel

Objectives of Remedial Action. In the event of an accident involving the release of significant quantities of radioactive material, the objectives of all remedial action are to:

a. Minimize the amount of radioactive material entering the body, by ingestion, inhalation, or through any wounds.
b. Prevent the spread of contamination from the area of the accident.
c. Remove radioactive contamination on personnel.
d. Start area decontamination procedures under qualified supervision. Inexperienced personnel should not attempt unsupervised decontamination.

Procedures for Dealing with Minor Spills and Contamination. Most accidents will involve only minor quantities or radioactivity (i.e., at the microcurie level).

a. Put on gloves to prevent contamination of the hands. (Wash hands first if they are contaminated as a result of accident.)

b. Drop absorbent paper or cloth on the spill to limit spread of contamination.

c. Mark off contaminated area. Do not allow anyone to leave contaminated area without being monitored.

d. Notify the radiation protection office of the accident.

e. Start decontamination procedures as soon as possible. Normal cleaning agents should be adequate. Keep cleaning supplies to a minimum needed to do the job and place into sealed bags after use. Recommendations for difficult jobs may be found in the Radiological Health Handbook, listed in the References. Proceed from the outermost edges of the contaminated area inward, reducing systematically the area that is contaminated. (This principle may not apply in decontamination of highly radioactive areas, which would require supervision by a radiation protection specialist.)

f. Put all contaminated objects into containers to prevent spread of contamination.

g. Assign a person equipped with a survey meter to follow the work and watch for accidental spread of contamination.

Personnel Decontamination. If personnel contamination is suspected, first identify contaminated areas with a survey meter. Do not use decontamination methods that will spread localized material or increase penetration of the contaminant into the body; e.g., by abrasion of the skin. Decontamination of wounds should be accomplished under the supervision of a physician.

Irrigate any wounds profusely with tepid water and clean with a swab. Follow with soap or detergent and water (and gently scrubbing with a soft brush, if needed). Avoid the use of highly alkaline soaps (may result in fixation of the contaminant) or organic solvents (may increase skin penetration by contaminant).

Use the following procedures on intact skin:

a. Wet hands and apply detergent.

b. Work up good lather; keep lather wet.

c. Work lather into contaminated area by rubbing gently for at least 3 minutes. Apply water frequently.

d. Rinse thoroughly with lukewarm water (limiting water to contaminated areas).

e. Repeat above procedures several times, gently scrubbing residual contaminated areas with a soft brush, if necessary.

f. If the radiation level is still excessive, initiate more powerful decontamination procedures after consulting with the radiation protection office.

Reporting Radiation Accidents. Those accidents involving radiation must be reported to the Nuclear Regulatory Commission. Reports must include information such as number of individuals exposed, names of individuals exposed, level of exposure, nuclides involved, and/or the concentration of nuclides released. The NRC has established notification requirements such that if an accident meets the following criteria, it must be reported immediately:

a. Whole-body exposure of individual exceeds 25 rems.

b. Skin exposure exceeds 150 rems.

c. Exposure of hands, feet, ankles exceeds 375 rems.

d. Release of radioactive materials exceeds 5000 times the specified limits over a 24-hour period.

e. Loss of one or more working weeks due to radiation accident.

f. Damage to property in excess of $100,000.

If the accident does not meet the above levels but meets the criteria below, the accident must be reported within 24 hours.

a. Whole-body exposure in excess of 5 rems.

b. Skin exposure exceeding 30 rems.

c. Exposure of hands, feet, ankles exceeds 75 rems.

d. Release of radioactive materials exceeding 500 times the specified concentration limits over a 24-hour period.

e. Loss of one or more working days due to a radiation accident.

f. Damage to property in excess of $1,000.

Responsibilities of the Industrial Hygiene Engineer

In some instances, the industrial hygiene engineer has direct responsibility for radioactive sources. In fulfilling this responsibility, the following activities should be performed:

1. Comply with all government regulations.
2. Ensure adequate supervision and training of personnel working with radioactivity.
3. Maintain inventory of radioactive sources being used, including type and amount.
4. Review operational procedures to evaluate potential:
   a. personnel exposure;
   b. accidents causing a radiation release.
5. Periodically survey radioactive areas for radiation levels.
6. Evaluate laboratory facility for proper design, construction, shielding, posting, labeling, etc.
7. Design alternative plans for containment and decontamination in the event of an accident.
8. Monitor the disposal of wastes.
9. Supervise the monitoring systems, including personnel monitoring procedures.
10. Maintain accurate records, including personnel cumulative exposure, radiation surveys, instrument calibration, waste disposal, and radiation incidents.

Summary

The last three chapters have presented an overview of the theory, use, monitoring, and control and management of ionizing radiation. Because of the potential health hazard of ionizing radiation, it is essential that the industrial hygiene engineer minimize the potential personnel exposure when working with radiation sources. These chapters were not intended to provide the reader with a complete and comprehensive understanding of ionizing radiation, but the material presented was designed to provide an initial understanding of the concepts of ionizing radiation.